

The dashboard should consist of three levels:

## 1. NATIONAL ENSO STRATEGIC DASHBOARD

### Executive Coordination Group Dashboard

#### National ENSO Status Indicator

| Indicator                             | Current Status                      | Trend | Alert Level         |
|---------------------------------------|-------------------------------------|-------|---------------------|
| ENSO Phase                            | Neutral / Watch / El Niño / La Niña | ↑ ↓ → | Green / Amber / Red |
| Oceanic Niño Index (ONI)              | Value                               | Trend | Alert               |
| Southern Oscillation Index (SOI)      | Value                               | Trend | Alert               |
| Sea Surface Temperature (SST) Anomaly | Value                               | Trend | Alert               |
| Rainfall Anomaly                      | Value                               | Trend | Alert               |

#### National Security Risk Dashboard

| Sector             | Risk Level          | Trend |
|--------------------|---------------------|-------|
| Food Security      | Low / Medium / High | ↑ ↓   |
| Water Security     | Low / Medium / High | ↑ ↓   |
| Public Health      | Low / Medium / High | ↑ ↓   |
| Energy Security    | Low / Medium / High | ↑ ↓   |
| Economic Stability | Low / Medium / High | ↑ ↓   |
| Infrastructure     | Low / Medium / High | ↑ ↓   |
| Social Stability   | Low / Medium / High | ↑ ↓   |

#### Provincial Risk Heat Map

##### High Risk Provinces

| Province           | Drought Risk | Food Risk | Water Risk | Overall Rating |
|--------------------|--------------|-----------|------------|----------------|
| Enga               | High         | High      | Medium     | Red            |
| Western Highlands  | High         | Medium    | Medium     | Amber          |
| Southern Highlands | Medium       | Medium    | Medium     | Amber          |
| Gulf               | Medium       | Low       | Medium     | Amber          |

Interactive GIS mapping should be included.

## 2. ENSO INTELLIGENCE CELL DASHBOARD

### NSA Secretariat

#### Climate Indicators

| Indicator | Source    | Frequency |
|-----------|-----------|-----------|
| ONI       | NOAA      | Weekly    |
| SOI       | BOM       | Daily     |
| SST       | NOAA/NASA | Daily     |

|                          |           |        |
|--------------------------|-----------|--------|
| Rainfall Anomalies       | PNGNWS    | Daily  |
| Temperature Anomalies    | PNGNWS    | Daily  |
| Soil Moisture            | NASA      | Weekly |
| Vegetation Stress (NDVI) | NASA      | Weekly |
| River Levels             | Water PNG | Weekly |

#### Economic Indicators

| Indicator             | Lead Agency          |
|-----------------------|----------------------|
| Food Prices           | Treasury             |
| CPI Inflation         | BPNG                 |
| Fuel Stocks           | Energy Agencies      |
| Commodity Exports     | Treasury             |
| Transport Disruptions | Transport Department |

#### Human Security Indicators

| Indicator           | Lead Agency |
|---------------------|-------------|
| Malnutrition Cases  | Health      |
| Disease Outbreaks   | Health      |
| Water Access Stress | Water PNG   |
| Crop Failures       | DAL         |
| School Attendance   | Education   |

### 3. NATIONAL ENSO EARLY WARNING TRAFFIC LIGHT SYSTEM

#### GREEN – Routine Monitoring

- Neutral ENSO Conditions
- Monthly Meetings
- Normal Monitoring

#### AMBER – ENSO WATCH

- SST anomalies increasing
- ONI approaching threshold
- Fortnightly Meetings
- Cluster Readiness Activated

#### RED – ENSO ALERT

- El Niño or La Niña declared
- Weekly Meetings
- Provincial Preparedness Measures Activated

#### BLACK – NATIONAL EMERGENCY

- Severe impacts occurring
- Daily Coordination
- National Emergency Operations Activated

#### REAL-TIME NOAA / NASA INTEGRATION DISPLAY

The dashboard should contain a dedicated Global ENSO Monitoring Panel displaying feeds from the following agencies:

#### Primary International Monitoring Sources

| Agency                                                              | Monitoring Function                     |
|---------------------------------------------------------------------|-----------------------------------------|
| <a href="#">NOAA Climate Prediction Center</a>                      | Official ENSO Outlook                   |
| <a href="#">NOAA ENSO Diagnostic Discussion</a>                     | Monthly ENSO Forecast                   |
| <a href="#">NOAA Coral Reef Watch SST Monitor</a>                   | Sea Surface Temperature                 |
| <a href="#">NASA Earth Observatory</a>                              | Satellite Climate Monitoring            |
| <a href="#">NASA Worldview Satellite Platform</a>                   | Near Real-Time Global Satellite Imagery |
| <a href="#">World Meteorological Organization (WMO)</a>             | Global ENSO Updates                     |
| <a href="#">Australian Bureau of Meteorology (BoM) ENSO Outlook</a> | Pacific Regional Forecasts              |
| <a href="#">Japan Meteorological Agency (JMA) ENSO Monitoring</a>   | Western Pacific Monitoring              |

#### Recommended Dashboard Architecture

##### Strategic Layer

Chief Secretary → NSAC → NEC

##### Intelligence Layer

NSA ENSO Intelligence Cell

##### Technical Layer

PNG National Weather Service

#### Data Sources

- NOAA
- NASA
- WMO
- BoM
- JMA

- CEPA
- Water PNG
- NDC
- Treasury
- BPNG
- Health
- DAL

#### **Visualization Platform**

- ArcGIS Dashboard
- Microsoft Power BI
- Tableau
- Web-based National Climate Intelligence Portal

For the National ENSO Monitoring Taskforce, I recommend a **Microsoft Power BI Executive Dashboard** with four integrated pages designed for the Chief Secretary, NSA, PNGNWS, NSAC, NEC, and Cluster Leads.

NATIONAL ENSO EARLY WARNING AND CLIMATE INTELLIGENCE DASHBOARD  
Microsoft Power BI Design Framework

OWNER:

National Security Agency (NSA)

TECHNICAL LEAD:

Papua New Guinea National Weather Service (PNGNWS)

PLATFORM:

Microsoft Power BI Service

REPORTING FREQUENCY:

Real-Time / Daily / Weekly

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DASHBOARD PAGE 1

EXECUTIVE STRATEGIC OVERVIEW

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HEADER KPI CARDS

Current ENSO Status

[Neutral | Watch | El Niño | La Niña]

National Alert Level

[Green | Amber | Red | Black]

National Risk Rating

[Low | Medium | High | Critical]

Affected Population Estimate

Number of High-Risk Provinces

Forecast Period

(1 Month | 3 Month | 6 Month)

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NATIONAL RISK MATRIX

Food Security Risk

Water Security Risk

Public Health Risk

Economic Stability Risk

Infrastructure Risk

Energy Security Risk

## Social Stability Risk

Each displayed using:

- Traffic Light Indicators
  - Trend Arrows
  - Risk Scores
- 

## PROVINCIAL HEAT MAP

PNG GIS Map

Colour Coding

Green = Low Risk

Amber = Moderate Risk

Red = High Risk

Dark Red = Critical Risk

Selectable by Province

Displays:

- Population
  - Water Stress
  - Crop Loss Risk
  - Drought Severity
  - Food Security Status
  - Vulnerability Ranking
- 

## DASHBOARD PAGE 2

### ENSO CLIMATE INTELLIGENCE

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## LIVE ENSO INDICATORS

Oceanic Niño Index (ONI)

Southern Oscillation Index (SOI)

Sea Surface Temperature Anomaly (SST)

Rainfall Anomaly

Temperature Anomaly

Soil Moisture Index

Vegetation Health Index

Displayed as:

- Gauge Charts
  - Trend Lines
  - Forecast Curves
- 

## ENSO TREND ANALYSIS

Last 12 Months

Forecast Next 6 Months

Historical Comparison

El Niño 1997–98

El Niño 2015–16

El Niño 2023–24

Current Event

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## EARLY WARNING PANEL

Trigger Thresholds

Green = Normal

Amber = Watch

Red = Alert

Black = Emergency

Automatic Escalation Notifications

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DASHBOARD PAGE 3

SECTORAL IMPACT MONITORING

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## FOOD SECURITY PANEL

Crop Conditions

Food Price Trends

Agricultural Production

Provincial Food Stress Index

Lead Agency:

Department of Agriculture & Livestock

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## WATER SECURITY PANEL

Reservoir Levels

River Levels

Water Demand

Urban Water Supply Status

Lead Agencies:

Water PNG

Eda Ranu

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#### PUBLIC HEALTH PANEL

Disease Outbreak Alerts

Malnutrition Cases

Waterborne Disease Monitoring

Lead Agency:

National Department of Health

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#### ECONOMIC SECURITY PANEL

Inflation

Fuel Availability

Commodity Production

Export Performance

Supply Chain Disruptions

Lead Agencies:

Treasury

Finance

BPNG

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DASHBOARD PAGE 4

ENSO INTELLIGENCE & OPERATIONS CENTRE

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#### NSA ENSO INTELLIGENCE CELL

National Situation Picture

Current Threat Assessment

High-Risk Provinces

Critical Infrastructure Status



Security Incidents

Humanitarian Operations

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## TASKFORCE OPERATIONS TRACKER

Cluster Status

Scientific & Data Cluster

Economic & Infrastructure Cluster

Human Security Cluster

Water & Environment Cluster

Security & Logistics Cluster

Status:

Active / Standby

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## ACTION TRACKER

Agency

Task

Due Date

Status

Responsible Officer

Progress %

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## STRATEGIC REPORTING PANEL

Automatic Generation Of:

Weekly ENSO SITREP

Monthly ENSO Intelligence Brief

Quarterly NSAC Brief

Quarterly NEC Brief

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## EXTERNAL DATA CONNECTIONS

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NOAA

Oceanic Niño Index

ENSO Outlook

Sea Surface Temperature

NASA

Satellite Imagery

NDVI Vegetation Index

Soil Moisture Data

Earth Observation Layers

BOM Australia

SOI

ENSO Outlook

JMA Japan

Pacific ENSO Monitoring

PNG National Weather Service

Rainfall Data

Temperature Data

Forecast Models

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## POWER BI TECHNOLOGY STACK

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Data Sources

Microsoft Excel

CSV

SQL Server

ArcGIS Online

NOAA APIs

NASA EarthData APIs

PNGNWS Data Feeds

Power BI Components

Power Query

Power BI Desktop

Power BI Service

Power BI Mobile

ArcGIS Maps for Power BI

Power Automate Alerts

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## EXECUTIVE REPORTING FLOW

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NOAA / NASA / BOM / JMA



PNG National Weather Service



NSA ENSO Intelligence Cell



ENSO Taskforce



Chief Secretary



NSAC



NEC

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## STRATEGIC OUTCOME

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The Power BI Dashboard will serve as Papua New Guinea's National ENSO Early Warning and Climate Intelligence Platform, integrating scientific monitoring, risk intelligence, sectoral preparedness, and executive decision support into a single national operating picture for Government.

### Recommended Visual Layout

#### Top Banner

- ENSO Status (Large Card)
- National Alert Level (Traffic Light)
- Forecast Outlook
- Date/Time of Last Update

#### Middle Section

- PNG Provincial Risk Heat Map
- ENSO Trend Graph (ONI, SOI, SST)
- National Risk Matrix

#### Bottom Section

- Cluster Status Board
- Critical Alerts
- Action Tracker
- Executive Recommendations

a **three-tier Power BI + ArcGIS National Climate Intelligence Architecture**. This configuration is suitable for deployment within the NSA Secretariat as the proposed ENSO Intelligence Cell and can later evolve into a National Climate Intelligence Centre.

## NATIONAL ENSO EARLY WARNING AND CLIMATE INTELLIGENCE SYSTEM (NEWCIS)

### POWER BI + ARCGIS ENTERPRISE DASHBOARD ARCHITECTURE

#### OWNER:

National Security Agency (NSA)

#### TECHNICAL LEAD:

Papua New Guinea National Weather Service (PNGNWS)

#### SUPPORTING AGENCIES:

NDC, DNPM, Finance, Treasury, Health, DAL, CEPA, Water PNG, PNGDF, RPNGC

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## SYSTEM ARCHITECTURE

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### GLOBAL DATA SOURCES

(NOAA, NASA, WMO, BOM, JMA)



### PNG NATIONAL WEATHER SERVICE



### NSA ENSO INTELLIGENCE CELL



### POWER BI DATA MODEL



### ARCGIS ENTERPRISE PLATFORM



### EXECUTIVE DASHBOARDS



Chief Secretary

NSAC

NEC

ENSO Taskforce

Cluster Leads

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## ARCGIS NATIONAL OPERATING PICTURE

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### BASE LAYER

PNG National Boundary

Provincial Boundaries

District Boundaries

LLG Boundaries

Road Network

Ports

Airports

Power Infrastructure

Water Infrastructure

Health Facilities

Schools

Telecommunications Infrastructure

Critical National Infrastructure

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## CLIMATE INTELLIGENCE LAYERS

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### Layer 1

Rainfall Anomalies

Source:  
PNGNWS

Update:  
Daily

Colour Scale:  
Blue → Normal  
Yellow → Dry  
Red → Severe Dry

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### Layer 2

Drought Severity Index

Source:  
NASA  
PNGNWS

Update:  
Weekly

Classification

Normal

Moderate

Severe

Extreme

Exceptional

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Layer 3

Sea Surface Temperature Anomalies

Source:

NOAA

Update:

Daily

Coverage

Pacific Ocean

PNG EEZ

Coral Sea

Bismarck Sea

Solomon Sea

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Layer 4

Vegetation Stress (NDVI)

Source:

NASA MODIS

Update:

Weekly

Displays

Agricultural Stress

Forest Stress

Crop Failure Risk

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Layer 5

Soil Moisture

Source:  
NASA SMAP

Update:  
Daily

Displays

Agricultural Water Availability

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Layer 6

River Flow Monitoring

Source:  
Water PNG

Update:  
Weekly

Displays

Low River Conditions

Critical Water Supply Risk

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HUMAN SECURITY LAYERS

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Food Security Index

Crop Production Status

Malnutrition Hotspots

Disease Outbreak Locations

Water Access Vulnerability

Population Exposure Estimates

Disaster Relief Distribution Sites

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ECONOMIC SECURITY LAYERS

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Agricultural Production Zones

Mining Operations

Petroleum Facilities

Power Generation Assets

Transport Corridors



Ports

Airports

Fuel Depots

Supply Chain Nodes

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## SECURITY LAYERS

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PNGDF Bases

Police Commands

Emergency Response Centres

Strategic Infrastructure

Critical National Assets

Emergency Logistics Routes

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## POWER BI EXECUTIVE DASHBOARD

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### PAGE 1

National ENSO Overview

Current ENSO Phase

National Alert Level

Forecast Outlook

National Risk Index

Population At Risk

High Risk Provinces

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### PAGE 2

Climate Intelligence Dashboard

ONI

SOI

SST

Rainfall Anomalies

Temperature Anomalies

Drought Severity

Forecast Trends

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PAGE 3

Sectoral Risk Dashboard

Food Security

Water Security

Public Health

Economic Security

Infrastructure Security

Energy Security

---

PAGE 4

Provincial Risk Dashboard

Interactive Province Selection

Risk Score

Vulnerability Assessment

Preparedness Status

Resource Requirements

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PAGE 5

Executive Decision Support Dashboard

Strategic Recommendations

Active Alerts

Cluster Status

Outstanding Actions

Preparedness Level

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AUTOMATED ALERT SYSTEM

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GREEN

Routine Monitoring

## Monthly Reports

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AMBER

ENSO Watch

Automatic Email Alert

Taskforce Notification

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RED

ENSO Alert

Immediate Notification

Weekly Meetings Triggered

Cluster Activation Triggered

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BLACK

National Emergency

Immediate Escalation

Chief Secretary Alert

NSAC Notification

NEC Notification

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## REPORTING PRODUCTS

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Weekly

ENSO Situation Report (SITREP)

Monthly

ENSO Strategic Intelligence Assessment

Quarterly

National Climate Security Assessment

Biannual

National ENSO Preparedness Report

Annual

Climate Security Outlook Report

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## FUTURE ENHANCEMENTS

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### Phase 1 (2026)

Power BI + ArcGIS Dashboard

### Phase 2 (2027)

National Climate Intelligence Portal

### Phase 3 (2028)

AI-Powered Risk Forecasting

### Phase 4 (2029-2030)

National Climate Intelligence Centre

Integrated Multi-Hazard Monitoring  
(ENSO, Drought, Floods, Cyclones,  
Landslides, Food Security)

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## STRATEGIC OUTCOME

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A single National Climate Intelligence Operating Picture providing real-time visibility of ENSO risks, provincial vulnerabilities, critical infrastructure exposure, and national preparedness status to support evidence-based decision-making by the Chief Secretary, NSA, NSAC, NEC, and the National ENSO Monitoring Taskforce.

### **Recommended Executive Screen Layout (Large Operations Centre Display)**

#### **Top Row**

- ENSO Status (Green/Amber/Red)
- ONI
- SOI
- SST Anomaly
- Forecast Outlook

#### **Centre Screen (60% of display)**

- ArcGIS Interactive Map of Papua New Guinea
- Drought Hotspots
- Rainfall Anomalies
- Water Stress Areas
- Food Security Risk Areas

- Critical Infrastructure

#### **Right Panel**

- Provincial Risk Rankings
- Active Alerts
- Cluster Activation Status

#### **Bottom Panel**

- NSA Strategic Assessment
- Recommended Actions
- Meeting Schedule
- Reporting Deadlines